Practice assignment 10 - Not Graded

This assignment will not be graded and is only for practice.

1 point

Match the functions of two variables in Column A with their directions in which the directional derivatives at (0,0) is maximized given in Column B.

- [ ] $(1) \longrightarrow (v)$
- [ ] $(1) \longrightarrow (iii)$
- [ ] $(2) \longrightarrow (iv)$
- [ ] $(3) \longrightarrow (iv)$
- [ ] $(2) \longrightarrow (vi)$
- [ ] $(4) \longrightarrow (v)$
- [ ] $(3) \longrightarrow (vi)$
- [ ] $(4) \longrightarrow (i)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(1) \longrightarrow (iii)$

$(2) \longrightarrow (iv)$

$(4) \longrightarrow (v)$

1 point

Which of the following statements is(are) correct?

- [ ] Maximum directional derivative at (0,0) for the function $f(x, y) = |x| + |y|$ is 1.
- [ ] The graph of the linear approximation to a function $f(x, y)$ at a given point P is the tangent plane to $f(x, y)$ at the point P.
- [ ] A tangent plane to a function $f(x, y)$ exists at every point.
- [ ] All directional derivatives of a function $f(x, y)$ at a point P are 0 if the tangent plane is parallel to the XY-plane.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The graph of the linear approximation to a function $f(x, y)$ at a given point P is the tangent

plane to $f(x, y)$ at the point P.
All directional derivatives of a function $f(x, y)$ at a point P are 0 if the tangent plane is parallel to the XY-plane.

1 point

Which of the following is(are) critical points of the scalar valued multivariable function $f(x, y) = 6x^2 - 2x^3 + 3y^2 - 6xy$?

- [ ] (1,1)
- [ ] (0,0)
- [ ] (1,-1)
- [ ] No critical points.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(1,1)
(0,0)

Consider two functions f_1, f_2 defined as

$$f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2)$$

and

$$f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2).$$

If $L_{f_1}(x, y, z)$ and $L_{f_2}(x, y, z)$ are the linear approximation of $f_1(x, y, z)$ and $f_2(x, y, z)$ at point (1, -1, 1), then find the value of $85(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z))$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 30.0

1 point

$z = f(x, y) = \frac{-7}{6}x^2 + \frac{1}{3}y^2 + \frac{11}{6}$ represents a surface. Let (x_0, y_0, z_0) be the point where the tangent plane to the given surface is parallel to the plane $7x - 8y + 3z + 2 = 0$. Find the value of $11x_0 + 5y_0 + 4z_0$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 55

1 point

Find the number of points where the tangent plane to the function $f(x, y) = x^3 + y^3 + 3x^2 - 9x - 12y + 20$ at the points is parallel to the XY-plane.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4

1 point

- Suppose $ax + by + cz = d_1$ is a tangent plane to the function $f(x, y) = x^3 + 3y^3 + 9xy$ at (-1,2) and $ax + by + cz = d_2$ is a tangent plane to the same function $f(x, y)$ at $(\alpha, 1)$.

  Use the above information to answer questions 7-8:

  1 point

  Which of the following options is(are) true?

  - [ ] $\alpha = -2$
  - [ ] $\alpha = 2$
  - [ ] There are only two possible candidates for α.
  - [ ] $|d_2 - d_1| = 12$

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  $\alpha = 2$
  $|d_2 - d_1| = 12$

  Find the value of $\frac{2a+b+c}{d_1+d_2}$.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  (Type: Numeric) 1

  1 point

- Let $L(x, y)$ be the linear approximation to $f(x, y) = xe^y + ln(x + y)$ at point (1,0).

  Use the above information to answer questions 9-10:

  Suppose $L(x, y) = ax + by + c$, where $a, b, c \in R$. Find the value of $a + b - c$.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  (Type: Numeric) 5

  1 point

  Find the approximate value of the function $f(x, y)$ at (1.1,0.1) using the linear approximation function $L(x, y)$.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  (Type: Numeric) 1.4

  1 point

  Check Answers

  Your score is: 0/10